CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-501: Introduction to DBMS

Session: December 2023 Examination | **Paper Code:** 18021 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) Define Database Management System (DBMS).

Definition: A **Database Management System (DBMS)** is a computerized system software suite that enables users to define, create, maintain, query, and control access to a shared database. It serves as an intelligent interface between end-user applications and physical data storage.

Q2 (3M) List any five applications of DBMS.

Applications:

1. **Banking:** Customer accounts, loans, ATM transactions, fund transfers.
2. **Airlines & Railways:** Real-time reservation systems, flight schedules, ticketing.
3. **Universities & Colleges:** Student admissions, course registrations, grades, fee accounts.
4. **E-Commerce & Retail:** Online product catalogs, shopping carts, billing, inventory control.
5. **Telecommunications:** Call records, monthly billing, subscriber information, network routing.

Q3 (3M) What are the disadvantages of a File Processing System?

Disadvantages:

- **Data Redundancy & Inconsistency:** Same data stored in multiple files leading to conflicting copies.
- **Difficulty in Accessing Data:** Requires writing a new custom program for every ad-hoc query.
- **Data Isolation:** Data scattered across disparate file formats cannot be easily joined.
- **Integrity Problems:** Hard to enforce consistency constraints across independent files.
- **Atomicity Failures:** Incomplete updates during crashes leave data in corrupt states.

Q4 (3M) Define Data Model.

Definition: A **Data Model** is an abstract framework providing a collection of conceptual concepts for describing data, relationships, semantics, and constraints in a database. Major categories: Conceptual/High-level (ER Model), Representational/Logical (Relational Model), and Physical/Low-level models.

Q5 (3M) What are Attributes? Give examples.

Definition: An **Attribute** is a descriptive property or characteristic that describes an entity in an entity set.
Examples for Entity STUDENT:

- *Simple Attribute:* `RollNo`, `Age`
- *Composite Attribute:* `FullName` (composed of `FirstName`, `LastName`)
- *Multivalued Attribute:* `PhoneNumbers`, `Hobbies`
- *Derived Attribute:* `Age` (computed from `DateOfBirth` and current date)

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) **Compare Generalization and Specialization with examples.**

Comparison:

Criteria	Generalization	Specialization
Approach	**Bottom-up** design process	**Top-down** design process
Mechanism	Combines common features of multiple lower-level entities into a single higher-level entity.	Splits a higher-level entity into specialized sub-entities based on distinct features.
Goal	Minimizes redundancy by factoring out common attributes.	Identifies distinguishing characteristics of specific sub-categories.
Example	Entities `CAR` and `TRUCK` generalized into higher entity `VEHICLE`.	Entity `EMPLOYEE` specialized into `SALARIED_EMP` and `HOURLY_EMP`.

Q7 (7.5M) **Explain Primary Key, Super Key, and Foreign Key with examples.**

Definitions & Relationships:

- **Super Key:** A set of one or more attributes that uniquely identifies a tuple within a relation. May contain redundant attributes (e.g., `{EmpID, Name}`).
- **Primary Key:** A strictly minimal candidate key selected by the database designer to uniquely identify each tuple. No duplicates, no `NULL` allowed.
- **Foreign Key:** An attribute that establishes a referential relationship by referencing the primary key of another table.

```
Relation: EMPLOYEE(EmpID, EmpName, Email, DeptNo)
Relation: DEPARTMENT(DeptNo, DeptName, Building)

Super Keys: {EmpID}, {EmpID, EmpName}, {Email}, {EmpID, Email}
Candidate Keys: {EmpID}, {Email}
Primary Key: {EmpID}
Foreign Key: {DeptNo} in EMPLOYEE references DeptNo in DEPARTMENT
```

Q8 (7.5M) **What is Normalization? Explain the importance of Normalization in database design.**

Definition: Normalization is the systematic process of organizing relation schemas based on functional dependencies to eliminate anomalies and avoid redundant storage.

Importance:

- Eliminates Insertion, Deletion, and Modification anomalies.
- Reduces storage space by ensuring each fact is recorded in only one place.
- Prevents data corruption by enforcing referential integrity constraints across smaller, well-structured tables.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

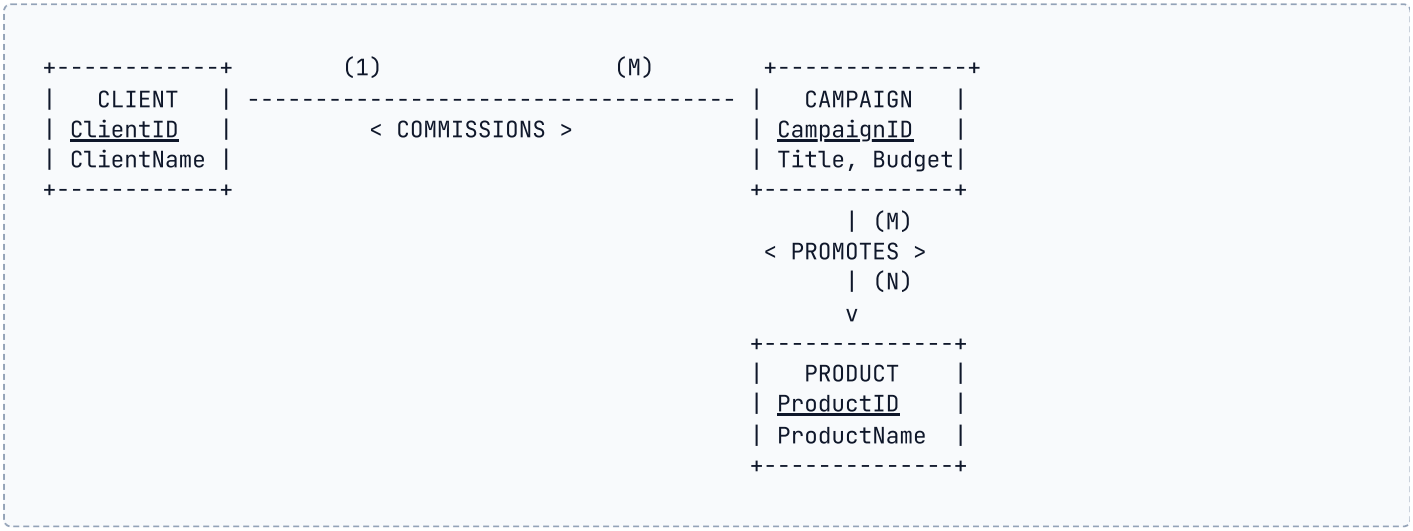
Q9 (15M) **Draw an ER Diagram for a small Marketing Company database, assuming your own data requirements.**

Enterprise Requirements:

- **CAMPAIGN:** CampaignID, Title, Budget, StartDate, EndDate.
- **CLIENT:** ClientID, ClientName, ContactPerson, Email, Phone.
- **PRODUCT:** ProductID, ProductName, Category.
- **MARKETING_AGENT:** AgentID, AgentName, CommissionRate.

Relationships:

- A Client commissions one or more Campaigns (1:N).
- Each Campaign promotes one or more Products (M:N).
- Agents are assigned to Campaigns (M:N).



Q10 (15M) **Describe the classification of Database Languages. Which type of language is SQL?**

Classification of Database Languages:

1. **Data Definition Language (DDL):** Specifies the database schema, storage structure, and constraints (`CREATE` , `ALTER` , `DROP` , `TRUNCATE`).
2. **Data Manipulation Language (DML):** Retrieves, inserts, updates, and deletes stored data. Can be procedural (relational algebra) or non-procedural/declarative (relational calculus, SQL).
3. **Data Control Language (DCL):** Controls permissions and access authorization (`GRANT` , `REVOKE`).
4. **Transaction Control Language (TCL):** Manages the logical unit of work (`COMMIT` , `ROLLBACK` , `SAVEPOINT`).

Classification of SQL: SQL is a **comprehensive declarative (non-procedural) language** that integrates DDL, DML, DCL, and TCL in a single standardized unified language.

Q11 (15M) **Explain DDL Commands and DML Commands with complete SQL syntax and executable examples.**

1. DDL Commands (Data Definition Language):

```
-- 1. CREATE TABLE
CREATE TABLE Student (
    RollNo INT PRIMARY KEY,
    Name VARCHAR(50) NOT NULL,
    Branch VARCHAR(20) DEFAULT 'BCA',
    FeePaid DECIMAL(10,2) CHECK (FeePaid ⩾ 0)
);

-- 2. ALTER TABLE: Add and modify column
ALTER TABLE Student ADD COLUMN Email VARCHAR(100);
ALTER TABLE Student MODIFY COLUMN Name VARCHAR(80);

-- 3. DROP & TRUNCATE
TRUNCATE TABLE Student; -- Deletes all rows, preserves table structure
DROP TABLE Student;     -- Deletes rows and schema permanently
```

2. DML Commands (Data Manipulation Language):

```
-- 1. INSERT
INSERT INTO Student (RollNo, Name, Branch, FeePaid, Email)
VALUES (101, 'Aman Sharma', 'BCA', 25000.00, 'aman@example.com');

-- 2. UPDATE
UPDATE Student
SET FeePaid = 30000.00
WHERE RollNo = 101;

-- 3. DELETE
DELETE FROM Student WHERE RollNo = 101;

-- 4. SELECT
SELECT RollNo, Name, FeePaid FROM Student WHERE Branch = 'BCA';
```

Q12 (15M) **Explain Data Independence with its types: Physical and Logical Data Independence. Why is it significant?**

Definition: Data Independence is the ability to modify schema definitions at one level of the 3-level ANSI/SPARC architecture without altering schemas at the higher level.

1. Logical Data Independence:

- The ability to modify the **Conceptual Schema** (e.g., adding/deleting attributes, splitting tables) without having to rewrite external schemas or existing application programs.
- *How achieved:* Using SQL Views that present the original logical schema to old applications.
- *Difficulty:* Harder to achieve because applications are closely tied to logical data models.

2. Physical Data Independence:

- The ability to modify the **Internal/Physical Schema** (e.g., changing file organizations, building B+ Tree indexes, moving to new storage media) without changing the Conceptual Schema.
- *Ease:* Easier to achieve because logical structure remains completely decoupled from disk-level storage.

Q13 (15M) Who are Data Administrators (DA)? What are the functions of a Database Administrator (DBA)? Differentiate between DA and DBA.

Data Administrator (DA) vs Database Administrator (DBA):

Aspect	Data Administrator (DA)	Database Administrator (DBA)
Role Level	Managerial and business level	Technical and operational level
Focus	Business data policies, enterprise data dictionary, data governance.	Physical database design, database server installation, query tuning, security.
Scope	Decides *what data* needs to be collected across the enterprise.	Decides *how data is stored, indexed, secured, and recovered*.

Key Functions of DBA:

1. Hardware/software installation and database upgrades.
2. Creating database users, granting roles, and monitoring access privileges.
3. Performance tuning: Analyzing execution plans, optimizing SQL queries, re-indexing.
4. Automating database backups and executing disaster recovery drills.